Curriculum Vitae - Kai-Chi Chang

Senior Research Scientist, University of California, Los Angeles: uclakcchang@g.ucla.edu

Research Interests

Quantum and Nonlinear Optics with Artificial Intelligence, High-Dimensional Photonic Quantum System, Quantum Information Science and Technology, Large-Alphabet Quantum Network and Communication Links, Chip-Scale Quantum Computing.

Professional Position and Activities

University of California, Los Angeles Jun. 2022 – Present
Senior Research Scientist Los Angeles, USA

- Nominee for UCLA Lecturer for Electrical and Computer Engineering 270 Applied Quantum Mechanics Fall 2025.
- IOP Trusted Reviewer Status, Feb. 2026.
- Advisor: Dr. Chee Wei Wong (Tannas Professor of Engineering).
- Review Scientific Journals: *Advanced Science*, *npj Quantum Information*, *Laser Photonics and Reviews*, *IEEE Journal on Selected Areas in Communications*, *IEEE Transactions on Information Forensics and Security*, *Quantum Science and Technology*, *APL Quantum*, *Frontiers of Physics*, *Optics Letters*, *Optics Express*, *Scientific Reports*, *New Journal of Physics*, *Review of Scientific Instruments*, *Chinese Journal of Physics*, *Physica Scripta*, *Journal of Lightwave Technology*, *IEEE Access*, *Optical and Quantum Electronics*, *Entropy*, *Optics Communication*s, *Brazilian Journal of Physics*, *Japanese Journal of Applied Physics*, and *IEEE Photonics Journal*.
- Review Scientific Education Textbook: *Hilbert Spaces with Applications*, *by Lokenath Debnath and Piotr Mikusinski*, *4th edition*, *Understanding Hilbert's Tenth Problem, by Domenico Cantone, Marianna Nicolosi-Asmundo*, *and Eugenio G. Omodeo*, *Provisional version*.

Education and Experiences

University of California, Los Angeles Sept. 2016 – June. 2022
Doctor of Philosophy in Electrical and Computer Engineering Los Angeles, USA

- PH.D. Dissertation: *High-Dimensional Quantum Information Processing with Time-Frequency Qudits.*
- Thesis committee: Prof. Jeffrey H. Shapiro, Prof. Andrei Faraon, Prof. Vwani Roychowdhury, Prof. Eric Hudson, and Prof. Chee Wei Wong.
- Academic Positions: Graduate Student Researcher, 2017-2021; Teaching Assistant, Electrical and Computer Engineering 170A Principle of Photonics Fall 2018, and Physics 1A Mechanics for Scientists and Engineers Winter 2020.
- GPA: 3.83/4.00.

National Chiao Tung University Feb. 2014 – May. 2016
Master of Science in Electrophysics Hsinchu, Taiwan

- Master Thesis: *Investigation of Subthreshold Emission Spectrum and Geometric Modes in Broad-Area Square Vertical-cavity Surface-emitting Laser with Resonant Model of Open Systems.*
- Supervisor: Prof. Yung-Fu Chen.

- GPA: 4.03/4.30.

National Central University Sept. 2010 – Jan. 2014
Bachelor of Science in Physics Taoyuan, Taiwan

- GPA Major: 3.93/4.00; GPA Overall: 3.92/4.00.
- Summer Research Intern at Institute of Atomic and Molecular Sciences, Academia Sinica, 2013.

Honors and Awards

- UCLA Chancellor Award for Postdoctoral Research Nominee (Top 3%), 2024.
- UCLA PH.D. Dissertation Research Award Finalist, 2022.
- UCLA Dissertation Year Fellowship (Top 2%), 2021-2022.
- UCLA J. Yang Scholarship, 2020-2021.
- APS FGSA URM Meeting Award Winner, 2021.
- SPIE Optics & Photonics Career Summit Officer Travel Grant Winner, 2020.
- Ministry of Education Fellowship, 2018-2020.
- UCLA Electrical and Computer Engineering Department Fellowship, 2016-2017.
- Honorary Member of The Phi Tau Phi Scholastic Honor Society of The Republic of China (Top 3%), 2016.
- Academic Excellence Award (Top 5%), 2010-2013, and 2015.

Publications Summary

14 publications (8 more manuscripts under review, including 1 at Science Advances, 1 at Physical Review X, 1 at Quantum), with h-index of **11**, including papers in Nature Photonics (x1), Newton (x1), npj Quantum Information (x1), Optica Quantum (x1), Photonics Research (x1), Communications Physics (x3), Quantum Science and Technology (x1), APL Quantum (x1), and Optics Express (x2). **46** Conferences papers in CLEO (including a highly-selected CLEO Postdeadline), APS and SPIE.

Journal Publications

***Equally contributing, joint first author ⁺Corresponding author**

1. **K.-C. Chang***,⁺, M. C. Sarihan, N. K. H. Li, F. Kanitschar, K. E. Akyüz, Y. Chen, D.-I. Lee, J. H. Kang, A. Aldhafeeri, A. Mueller, M. D. Shaw, B. Korzh, P. Erker, M. Spiropulu, M. Huber, and C. W. Wong, Unlocking the potential of high-dimensional quantum communication with scalable photonic entanglement in time and frequency, under review at *Sci. Adv.* (2026) (minor revision).

2. **K.-C. Chang***,⁺, M. C. Sarihan, P. Erker, X. Cheng, N. K. H. Li, K. E. Akyüz, A. Mueller, M. D. Shaw, B. Korzh, M. Spiropulu, M. Huber, and C. W. Wong, Measurement-efficient assumption-free certification of high-dimensional qudit steering and entanglement in time-frequency bases, under review at *Phy. Rev. X* (2026) (under revision). Preprint available as arXiv:2310.20694v3 (2025).

3. K. E. Akyüz, **K.-C. Chang***,⁺, C. Fan, Y. Chen, and C. W. Wong, Optimum adaptive multi-user high-dimensional quantum network, under preparation (2026).

4. **K.-C. Chang**[*,+], K. E. Akyüz, N. Wang, Y. Chen, and C. W. Wong, Resilience of qudits in hybrid classical-quantum secure frequency-multiplexed communication channels, under review (2026).

5. **K.-C. Chang**[*,+], M. C. Sarihan, K. E. Akyüz, L. Tauz, M. D. Shaw, B. Korzh, L. Dolecek, and C. W. Wong, High-rate and non-binary error-corrected quantum links via entangled photons, under review at *Commun. Eng.* (2026).

6. M. C. Sarihan, **K.-C. Chang**[*,+], X. Cheng, A. Shlosberg, C. Fan, G. Smith, and C. W. Wong, Resilient multi-node *d*-dimensional networks for high-rate entanglement distribution and quantum communications, under review at *npj Quan. Inf.* (2026).

7. Y. Chen, X. Cheng, **K.-C. Chang**[*,+], and C. W. Wong, Spatial quantum beating in quantum frequency combs, under preparation (2026).

8. **K.-C. Chang**[*,+], X. Cheng, Y. Chen, and C. W. Wong, Asymmetric photon bunching recurring interferences and distribution via quantum frequency combs, under review at *New J. Phys.* (2026).

9. Y. Chen, X. Cheng, **K.-C. Chang**[*,+], and C. W. Wong, Discretized frequency-bin entanglement via polarization manipulation, under preparation (2026).

10. **K.-C. Chang**[*,+], A. Mirani, M. C. Sarihan, X. Cheng, M. Harasimowicz, P. Hayden, and C. W. Wong, GKP-inspired high-dimensional superdense coding with energy-time entanglement, under review at *Quan.* (2026).

11. Y. Chen, X. Cheng, **K.-C. Chang**[*,+], and C. W. Wong, High-dimensional quantum key distribution via dual-basis interferometry, under preparation (2026).

12. **K.-C. Chang**[*,+], X. Cheng, M. C. Sarihan, and C. W. Wong, Recent advances in high-dimensional quantum frequency combs, *Newton* **1**, 100024 (2025).

13. **K.-C. Chang**[*,+], X. Cheng, F. R.-Hirsch, M. C. Sarihan, Y. Chen, J. G. F. Flores, M. Yu, P. G.-Q. Lo, D.-L. Kwong, and C. W. Wong, Quantum teleportation of a silicon nanophotonic CNOT gate, *Optica Quan.* **3**, 381 (2025).

14. **K.-C. Chang**[*,+], M. C. Sarihan, X. Cheng, Z. Zhang, and C. W. Wong, Large-alphabet time-bin quantum key distribution and Einstein-Podolsky-Rosen steering via dispersive optics, *Quan. Sci. Technol.* **9**, 015018 (2024).

15. **K.-C. Chang**[*,+], X. Cheng, M. C. Sarihan, and C. W. Wong, Time-reversible and fully time-resolved ultranarrow-band biphoton frequency combs, *APL Quan.* **1**, 016106 (2024).

16. X. Cheng, **K.-C. Chang**[*,+], M. C. Sarihan, A. Mueller, M. Spiropulu, M. D. Shaw, B. Korzh, A. Faraon, F. N. C. Wong, J. H. Shapiro, and C. W. Wong, High-dimensional time-frequency entanglement in a singly-filtered biphoton frequency comb, *Commun. Phys.* **6**, 278 (2023).

17. **K.-C. Chang**[*,+], X. Cheng, M. C. Sarihan, and C. W. Wong, Towards optimum Franson

interference recurrence in mode-locked singly-filtered biphoton frequency combs, *Photon. Res.* **11**, 1175 (2023).

18. X. Cheng, **K.-C. Chang*,+**, Z. Xie, M. C. Sarihan, Y. S. Lee, Y. Li, X. Xu, A. V. Kumar, S. Kocaman, M. Yu, P. G.-Q. Lo, D.-L. Kwong, J. H. Shapiro, F. N. C. Wong, and C. W. Wong, A chip-scale polarization-spatial-momentum quantum SWAP gate in silicon nanophotonics, *Nat. Photon.* **17**, 656 (2023).

19. **[Top 1% of all research outputs scored by altimetric]** **K.-C. Chang*,+**, X. Cheng, M. C. Sarihan, A. V. Kumar, Y. S. Lee, T. Zhong, Y.-X. Gong, Z. Xie, J. H. Shapiro, F. N. C. Wong, and C. W. Wong, 648 Hilbert space dimensionality in a biphoton frequency comb: entanglement of formation and Schmidt mode decomposition, *npj Quan. Inf.* **7**, 48 (2021).

20. X. Cheng, M. C. Sarihan, **K.-C. Chang***, Y. S. Lee, F. Laudenbach, Z. Yu, and C. W. Wong, Design of spontaneous parametric downconversion in integrated hybrid Si_xN_y-PPLN waveguides for high-dimension qubit entanglement, *Opt. Express* **27**, 30773 (2019).

21. Z. Xie, K.-H. Lou, **K.-C. Chang+**, N. C. Panoiu, H. Herrmann, C. Silberhorn, and C. W. Wong, Efficient C-band single-photon upconversion with chip-scale Ti-indiffused pp-$LiNbO_3$ waveguides, *Appl. Opt.* **58**, 5910 (2019).

22. S. Haldar, P. J. Barge, X. Cheng, **K.-C. Chang**, B. T. Kirby, S. Khatri, C. W. Wong, and H. Lee, Reducing classical communication costs in multiplexed quantum repeaters using hardware-aware quasi-local policies, *Commun. Phys.* **8**, 132 (2025).

23. S. Yang, M. C. Sarihan, **K.-C. Chang**, S.Venkatasubramanian, C. W. Wong, and L. Dolecek, Efficient non-binary spatially-coupled error correction codes for high-dimensional quantum communication channels, under preparation (2026).

24. J. Huang, A. Miranda, W. Liu, X. Cheng, B. Dwir, A. Rudra, **K.-C. Chang**, E. Kapon, and C. W. Wong, Spatial quantum-interference landscapes of exciton polaritons with multi-site-controlled quantum dots in extended cavity modes, *Commun. Phys.* **8**, 152 (2025).

25. C. Fan, M. C. Sarihan, J. Huang, J. H. Kang, **K.-C. Chang**, B. Liang, and C. W. Wong, Electron paramagnetic resonance and optical properties of a self-interstitial color center in silicon, under review at *Commun. Mat.* (2026).

26. Y. T. Yu, P. H. Tuan, **K.-C. Chang**, Y. H. Hsieh, Y. F. Chen, and K. F. Huang, Exploiting broad-area surface emitting lasers to manifest the path-length distributions of finite-potential quantum billiards, *Opt. Express* **24**, 82 (2016).

Summary of Scientific Accomplishments and Milestones

- Intensive collaboration with senior faculty (Prof. Jeffrey H. Shapiro and Dr. Franco N. C. Wong at MIT) and across other top institutions (Prof. Andrei Faraon and Prof. Maria Spiropulu at Caltech; Prof. Marcus Huber at Technical University Vienna; Prof. Tian Zhong at University of Chicago, Prof. Zheshen Zhang at University of Michigan, Ann Arbor, Prof. Graeme Smith at University of Colorado, Boulder, and Prof. Hayden at Stanford University), contributing to several manuscripts that have been either published or submitted to high-impact journals under several major funding support of National

Science Foundation (EFRI ACQUIRE 1741707, QII-TAQS 1936375, QuIC-TAQS 2137984), and Army Research Office (MURI W911NF-21-2-0214).

- Performed precise quantum optical measurements including nonlocal Franson interference, Hong-Ou-Mandel interferences, high-dimensional entanglement and Einstein-Podolsky-Rosen steering certification, and quantification using photonic qudits, which are useful for various quantum communication tasks.
- Successfully distributing high-dimensional energy-time entanglement, quantum steering and frequency-multiplexed quantum keys using quantum photonic resources.
- Achieved dual-chip distribution of polarization-entangled quantum states via high fidelity silicon polarization-spatial-momentum on-chip two-qubit SWAP gate, which can be used as a qubit converter in quantum internet.
- Demonstrated quantum teleportation of an elementary silicon photonic two-qubit CNOT gate towards distributed quantum computing.

Grants

- Assisted and won grants during postdoctoral research: National Science Foundation (QII-TAQS 1936375, $1,650,000, 5-years; QuIC-TAQS 2137984, $2,500,000, 4-years), and Army Research Office (MURI W911NF-21-2-0214, $3,750,000, 4-years).
- Assisted and won grants during PhD research: National Science Foundation (EFRI ACQUIRE 1741707, $2,000,000, 4-years).

Services and Outreaches

- Serving as a panelist in Quantum Research and Public Policy Workshop at UCLA, 2025.
- Serving as mentor in CLEO: IEEE Mentor Match program, 2024.
- Serving judge for student posters at IFoRE, 2023.
- Contributing to prepare course materials for Master of Quantum Science and Technology program at UCLA, 2023-2024.
- Mentor two PH.D., three master and four undergraduate UCLA students, 2018-2025.
- Serving vice-president at UCLA OSA and SPIE student chapters, 2018-2022.
- Invitation to external speakers from Universities and Industries in USA to UCLA for giving seminars on science, technology, and career developments, 2019-2022.
- Organized an explore-your-universe event at UCLA that focused on interesting and application optics for outreaching high-school students and kids, 2019.

Invited Talks

- ECE Department Seminar at Northeastern University, Feb. 2025.
- Physics and Astronomy Department Seminar at UCR, Apr. 2025.
- Quantum Network Science Seminar at UMass Amherst, Apr. 2025.
- Center of Quantum Networks Seminar at University of Arizona, Sep. 2025.
- Virtual Seminar at Photonics Group of University of Florida, Sep. 2025.
- SITP Seminar at Leinweber Institute for Theoretical Physics at Stanford University, Dec. 2025.
- ECE Department Seminar at SUNY Buffalo, Jan. 2026.
- The 15th Advanced Lasers and Photon Sources Conference (ALPS2026) of the Optics & Photonics International Congress 2026 (OPIC2026) at PACIFICO Yokohama, Japan, Apr. 2026.

- EntangleTalks of QCSA at UCLA, April. 2026.

Professional Skills

- Extensive experiences with local and nonlocal quantum interference measurements, femtosecond scale quantum precision measurements, high-dimensional entanglement certification, high-dimensional quantum communication with quantum error correction, on-chip quantum information processing with high-fidelity two-qubit gates, and design of high-purity hybrid chip-scale entangled photon source.
- Quantum photonic chip designs via COMSOL, Lumerical, and KLayout.
- Simulating Hamiltonian quantum systems with Qiskit and Qutip.
- Performing self-corrected quantum polarization algorithms using Quantum Optics Jena PAM devices.
- Realizing efficient error-correction in high-dimensional quantum communication testbeds using low-density parity check codes.
- Extracting information Shannon entropy and steering robustness in quantum photonic qudit platforms.

Conferences

***Presenter**

1. **K.-C. Chang***, K. E. Akyüz, N. Wang, Y. Chen, and C. W. Wong, "Noise robust certification of photonic qudit," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2026), paper FM3B.5 (oral).

2. **K.-C. Chang***, K. E. Akyüz, N. Wang, Y. Chen, and C. W. Wong, "Noise resilient wavelength-multiplexed high-dimensional quantum key distribution," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2026), paper FM4C.4 (oral).

3. **K.-C. Chang***, Y. Chen, K. E. Akyüz, N. Wang, M. Yu, P. G-Q. Lo, D.-L. Kwong, S. Kocaman, and C. W. Wong, "On-chip quantum nanophotonic spatial-polarization controlled-phase gate," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2026), paper JW1.122 (poster).

4. **K.-C. Chang***, M. C. Sarihan, P. Erker, X. Cheng, N. K. H. Li, K. E. Akyüz, A. Mueller, M. D. Shaw, B. Korzh, M. Spiropulu, M. Huber, and C. W. Wong, "High-dimensional entanglement witness using non-perfect correlations," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2026), paper JW1.216 (poster).

5. K. E. Akyüz*, C. Huang, **K.-C. Chang**, and C. W. Wong, "Convolution neural network for quantum network," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2026), paper FFxxx.x (oral).

6. **K.-C. Chang***, M. C. Sarihan, P. Erker, X. Cheng, N. K. H. Li, K. E. Akyüz, A. Mueller, M. D. Shaw, B. Korzh, M. Spiropulu, M. Huber, and C. W. Wong, "Qudit entanglement and steering certification in energy-time domain", American Physical Society March Meeting, SMT 2026 (oral).

7. K. E. Akyüz, **K.-C. Chang**, M. C. Sarihan, Y. Chen, C. Fan, X. Cheng, C. Huang, and C. W. Wong*, "High-dimensional Quantum Networks: deploying the frequency and time modalities,"

Photonics West (Quantum Computing, Communication, and Simulation VI), San Francisco, CA, January 2026 (oral).

8. **K.-C. Chang***, X. Cheng, F. R.-Hirsch, M. C. Sarihan, Y. Chen, J. G. F. Flores, M. Yu, P. G-Q. Lo, D.-L. Kwong, and C. W. Wong, "Quantum gate teleportation via a CMOS-integrated photonic chip," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2025), paper SS167_2 (oral).

9. **K.-C. Chang***, M. C. Sarihan, X. Cheng, P. Erker, A. Mueller, M. D. Shaw, B. Korzh, M. Spiropulu, M. Huber, and C. W. Wong, "Certification of high-dimensional quantum steering via experimentally-friendly energy-time measurements," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2025), paper FF105_5 (oral).

10. **K.-C. Chang***, A. Mirani, M. C. Sarihan, X. Cheng, M. Harasimowicz, P. Hayden, and C. W. Wong, "Superdense coding via time-frequency Gottesman-Kitaev-Preskill states", American Physical Society March Meeting, SMT 2025 (oral).

11. **K.-C. Chang***, X. Cheng, Y. Chen, K. E. Akyüz, H.-H. Chin, M. C. Sarihan, and C. W. Wong, "Distribution of photon bunching recurrences via a Quantum Frequency Comb," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2025), paper JPS200_18 (poster).

12. K. E. Akyüz*, **K.-C. Chang**, M. C. Sarihan, Y. Chen, X. Cheng, and C. W. Wong, "Adaptive entanglement distribution in wavelength-multiplexed, high-dimensional time-bin multi-user quantum networks," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2025), paper FF116_2 (oral).

13. Y. Chen, X. Cheng, M. C. Sarihan, **K.-C. Chang***, and C. W. Wong, "High-dimensional quantum key distribution secured by dual-basis interferometry," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2025), paper FF102_5 (oral).

14. X. Cheng, Y. Chen*, **K.-C. Chang**, and C. W. Wong, "Discrete frequency-bin entanglement generation via polarization-induced Hong-Ou-Mandel type interference," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2025), paper FF118_6 (oral).

15. **K.-C. Chang***, X. Cheng, Y. Chen, H.-H. Chin, M. C. Sarihan, S. C. Song, and C. W. Wong, "Asymmetric Hong-Ou-Mandel revivals interferences in a singly-filtered biphoton frequency comb," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2024), paper JW2A.136 (poster).

16. M. C. Sarihan, **K.-C. Chang***, Y. Chen, X. Cheng, H.-H. Chin, and C. W. Wong, "Enhancing High-Dimensional Arrival-time-bin Encoding through Polarization Hyperentanglement," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2024), paper JW2A.225 (poster).

17. X. Cheng*, **K.-C. Chang,** Y. Chen, M. C. Sarihan, and C. W. Wong, "Dual quantum comb generation via cascaded cavities," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2024), paper FTu4F.2 (oral).

18. Y. Chen*, X. Cheng, **K.-C. Chang,** M. C. Sarihan, S. C. Song, H.-H. Chin, and C. W. Wong, "Spatial quantum beating in a cavity-filtered biphoton frequency comb," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2024), paper FTu4F.5 (oral).

19. X. Cheng*, Y. Chen, **K.-C. Chang,** M. C. Sarihan, and C. W. Wong, "Deterministic polarization entanglement generation via wavelength division multiplexing for long distance distribution," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2024), paper JW2A.131 (poster).

20. H. H. Chin*, T. Melton, A. Aldhafeeri, L. Rukh, G. M. Colación, Y. Chen, X. Cheng, **K.-C. Chang**, T. E. Drake, and C. W. Wong, "Tuning of Carrier Envelope Offset Frequency in Octave-Spanning Si3N4 Microcomb," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2024), paper JW2A.17 (poster).

21. S. C. Song*, **K.-C. Chang**, X. Cheng, M. C. Sarihan, and C. W. Wong, "Optimal non-local Franson bi-photon quantum interferometry via high-finesse cavities in quantum communications," American Physical Society March Meeting, Vol. 2024, pp. EE02.00008 (oral).

22. M. C. Sarihan*, X. Cheng, **K.-C. Chang**, A. Shlosberg, C. Fan, G. Smith, and C. W. Wong, "A scalable multi-partite high-dimensional quantum network for high-rate entanglement distribution and secure communications," American Physical Society March Meeting, Vol. 2024 pp. Q53.00006 (oral).

23. **K.-C. Chang***, M. C. Sarihan, X. Cheng, and C. W. Wong, "Energy-time entangled and Einstein-Podolsky-Rosen steerable large-alphabet quantum key distribution," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2023), paper FF2A.7 (oral).

24. X. Cheng*, **K.-C. Chang**, M. C. Sarihan, and C. W. Wong, "Dispersion-managed Hong-Ou-Mandel revival via a biphoton frequency comb," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2023), paper FTu3A. 4 (oral).

25. **K.-C. Chang***, X. Cheng, M. C. Sarihan, and C. W. Wong, "Time-reversed ultranarrow-band telecommunication photon-pair sources over 10,000 modes," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2023), paper JTh2A.19 (poster).

26. M. C. Sarihan*, X. Cheng, **K.-C. Chang**, and C. W. Wong, "Wavelength-Multiplexed Multi-User Quantum Network Based on High-Dimensional Time-bin Encoding," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2023), paper FM1A.5 (oral).

27. M. C. Sarihan*, K. Yu, M. Taylor, A. Banerjee, J. H. Kang, C. Fan, **K.-C. Chang**, C. W. Wong, "Planar multi-mode superconducting circuit design for high-dimensional computation," American Physical Society March Meeting, Vol. 2023, pp. D75.00010 (oral).

28. **K.-C. Chang***, X. Cheng, M. C. Sarihan, W. Wang, F. N. C. Wong, J. H. Shapiro, and C. W. Wong, "Mode-locked phase coherent singly-resonant biphoton frequency comb," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2022), paper FTh5O.4 (oral).

29. X. Cheng*, Z. Xie, **K.-C. Chang**, M. C. Sarihan, Y. S. Lee, Y. Li, M. Yu, D.-L. Kwong, J.

H. Shapiro, F. N. C. Wong, and C. W. Wong, "A phase-coherent on-chip single-photon SWAP gate," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2022), paper FF3K.5 (oral).

30. M. C. Sarihan*, S. Yang, **K.-C. Chang**, S. Venkatasubramanian, L. Dolecek, and C. W. Wong, "Photon-efficient energy-time entanglement QKD using spatially-coupled irregular-repeat-accumulate error correction codes," American Physical Society March Meeting, Vol. 2022, pp. A35-008 (oral).

31. **[Invited Talk]** C. W. Wong, **K.-C. Chang***, X. Cheng, and M. C. Sarihan, 648-Hilbert space dimensionality in biphoton frequency combs for quantum-secure communications and networks, *Photonics West (AI and Optical Data Sciences III)*, San Francisco, CA, January 2022 (oral).

32. **[CLEO Postdeadline presentation]** **K.-C. Chang***, X. Cheng, M. C. Sarihan, F. N. C. Wong, J. H. Shapiro, and C. W. Wong, "High-dimensional energy-time entanglement distribution via a biphoton frequency comb," Conference on Lasers and Electro-Optics, OSA Technical Digest (Optical Society of America, 2021), paper FF1A.7 (oral).

33. **K.-C. Chang***, X. Cheng, M. C. Sarihan, A. V. Kumar, Y. S. Lee, T. Zhong, Y.-X. Gong, Z. Xie, J. H. Shapiro, F. N. C. Wong, and C. W. Wong, "Quantification of high-dimensional energy-time entanglement in a biphoton frequency comb," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2021), paper FM3M.6 (oral).

34. M. C. Sarihan*, **K.-C. Chang**, X. Cheng, Y. S. Lee, C. Chen, T. Zhong, H. Zhou, Z. Zhang, F. N. C. Wong, J. H. Shapiro, and C. W. Wong, "Proof-of-principle frequency-bin quantum key distribution with biphoton frequency combs," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2021) paper ATu1S.6 (oral).

35. X. Cheng*, M. C. Sarihan, **K.-C. Chang**, C. Chen, F. N. C. Wong, and C. W. Wong, "Secure high-dimensional quantum key distribution based on wavelength-multiplexed time-bin encoding," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2021), paper STh1D.3 (oral).

36. X. Cheng*, Z. Xie, **K.-C. Chang**, M. C. Sarihan, Y. S. Lee, A. K. Vinod, Y. Li, X. Xu, S. Kocaman, M. Yu, D.-L. Kwong, J. H. Shapiro, F. N. C. Wong, and C. W. Wong, "Quantum state tomography of an on-chip polarization-spatial qubit SWAP gate," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2021), paper FF2I.2 (oral).

37. **K.-C. Chang***, X. Cheng, M. C. Sarihan, A. V. Kumar, Y. S. Lee, T. Zhong, Y.-X. Gong, Z. Xie, J. H. Shapiro, F. N. C. Wong, and C. W. Wong, "648 Hilbert space dimensionality in a biphoton frequency comb," in American Physical Society March Meeting 2021, Abstract: H71.00140 (poster).

38. **K.-C. Chang***, X. Cheng, M. C. Sarihan, A. V. Kumar, Y. S. Lee, T. Zhong, Y. Gong, Z. Xie, J. H. Shapiro, F. N. C. Wong, and C. W. Wong, "High-dimensional time-frequency entanglement and Schmidt number witnesses using a biphoton frequency comb," Conference on Lasers and Electro-Optics (Optica Publishing Group, 2020), paper JTh2A.23 (poster).

39. M. C. Sarihan*, **K.-C. Chang**, X. Cheng, Y. S. Lee, C. Chen, T. Zhong, H. Zhou, Z. Zhang, F. N. C. Wong, J. H. Shapiro, and C. W. Wong, "Frequency-multiplexed rate-adaptive quantum key distribution with high-dimensional encoding," Conference on Lasers and Electro-Optics(Optica Publishing Group, 2020), paper FF3C.3 (oral).

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Work Authorization

- U.S. Permanent Resident (Green Card holder); no sponsorship required.